Product Requirements Document (PRD) – RidesRus

---

1. Overview

RidesRus is a smart bus-hailing platform designed for Lagos commuters. The platform allows passengers to see available buses in real time, hail specific buses, board securely via QR code, pay seamlessly, and select their drop-off stops. Drivers use a mounted tablet console for passenger management and trip coordination.

The system aims to modernize intra-city bus transportation by combining the efficiency of ride-hailing with the scalability of bus systems.

---

2. Objectives / Goals

- Provide **real-time bus visibility** with seat availability.
- Enable **secure, cashless boarding** via QR code.
- Simplify driver operations with a **console dashboard** (instead of phone apps).
- Ensure **automated passenger flow** from hail → boarding → stop → completion.
- Collect valuable **usage analytics** for route optimization and business decisions.

---

3. Key Assumptions

- Passengers will have smartphones with internet access.
- Each bus will have a mounted **Android tablet** with internet + GPS.
- Payment processing will rely on **Paystack/Quickteller/Flutterwave** (local gateways).

- GPS accuracy in Lagos may vary but will be acceptable with adjustments.
- Initial rollout: **6 buses between CMS and Adesanya**.

---

4. Core Features & Requirements

Feature 1: Live Bus Tracking & Seat Availability

Description: Passengers view buses on a map, color-coded by seat availability (white = empty, green = seats available, red = full).

- **Functional Requirement**: GPS tracking for each bus, seat count synced to server.
- **User Stories**:
    1. *As a passenger, I want to see buses near me so I know which one to board.*
    2. *As a passenger, I want to know if there are free seats before hailing.*
    3. *As an admin, I want to track buses to monitor service coverage.*
- **Success Metrics**: 90%+ of passengers use map to find buses daily.

Feature 2: Hailing a Bus

Description: Passenger taps a bus → pop-up → hail option → driver console alerted.

- **Functional Requirement**: Pop-up UI, hail request API, bus console notifications.
- **User Stories**:
    1. *As a passenger, I want to hail a bus so the driver knows to stop for me.*
    2. *As a driver, I want hail notifications so I don't miss passengers.*
    3. *As an admin, I want hail logs for demand analysis.*

- **Success Metrics**: 80% of hails successfully converted into boardings.

Feature 3: Boarding via QR Code

Description: Boarding only happens after passenger scans QR. This confirms hail, processes payment, and updates seat count.

- **Functional Requirement**: QR code generator per bus, in-app scanner, payment API.
- **User Stories**:
    1. *As a passenger, I want to scan a QR to confirm my booking and pay.*
    2. *As a driver, I want boarding to update automatically without manual work.*
    3. *As an admin, I want secure QR logs to prevent fraud.*
- **Success Metrics**: 95%+ QR scans successful on first try.

Feature 4: Stop Selection After Boarding

Description: Once seated, passenger selects drop-off stop. If app is closed, push notification prompts them.

- **Functional Requirement**: Stop selection UI, push notifications, driver console sync.
- **User Stories**:
    1. *As a passenger, I want to select my stop after boarding so it's convenient.*
    2. *As a driver, I want upcoming stops shown so I don't miss them.*
    3. *As an admin, I want stop data for route optimization.*
- **Success Metrics**: 90%+ passengers successfully select stops before drop-off.

Feature 5: Bus Console Dashboard

Description: Mounted tablet shows driver passenger count, seats left, upcoming stops, and hail requests.

- **Functional Requirement**: Tablet app with large fonts, minimal distractions, offline fallback.
- **User Stories**:
    1. *As a driver, I want a dashboard so I can easily see stops and passenger count.*
    2. *As a passenger, I want drivers to have accurate info so they don't miss me.*
    3. *As an admin, I want dashboards to sync live for monitoring operations.*
- **Success Metrics**: Drivers report <10% missed hails due to dashboard clarity.

Feature 6: Trip Completion & Feedback

Description: At drop-off, passenger sees a completion screen with thank-you message + feedback option.

- **Functional Requirement**: End-of-trip detection, feedback form, analytics storage.
- **User Stories**:
    1. *As a passenger, I want a trip completion message so I know my trip is done.*
    2. *As a driver, I want completed trips logged for accountability.*
    3. *As an admin, I want feedback to measure service quality.*
- **Success Metrics**: 30%+ passengers leave feedback after trips.

---

5. SWOT Analysis (Overall Product)

- **Strengths**:
    - First-to-market structured bus-hailing in Lagos.
    - Improves commuter trust and efficiency.
    - Fully digital, less cash handling.
- **Weaknesses**:
    - High reliance on internet and GPS.
    - Requires passenger smartphones and bus tablets.
    - Upfront hardware investment for buses.
- **Opportunities**:
    - Expand routes across Lagos & beyond.
    - Introduce corporate subscriptions (office shuttles).
    - Add loyalty/rewards system for frequent riders.
- **Threats**:
    - Competitors (Uber Bus, SWVL-style startups).
    - Regulatory pushback on new transport systems.
    - Infrastructure challenges (poor roads, network downtime).

---

6. Potential Issues / Risks

- **Tech Risks**: GPS inaccuracies, payment gateway downtime, app crashes.
- **Operational Risks**: Drivers ignoring console, passengers not scanning QR.
- **User Risks**: Smartphone penetration & adoption barriers.

- **Financial Risks**: High cost of scaling with hardware (tablets, mounts, internet).

---

7. Success Metrics (KPIs)

- **Adoption**: Number of active passengers per day.
- **Reliability**: % of successful boardings vs hails.
- **Revenue**: Total fare volume processed via app.
- **Efficiency**: Average waiting time at stops.
- **Satisfaction**: Average feedback rating per trip.